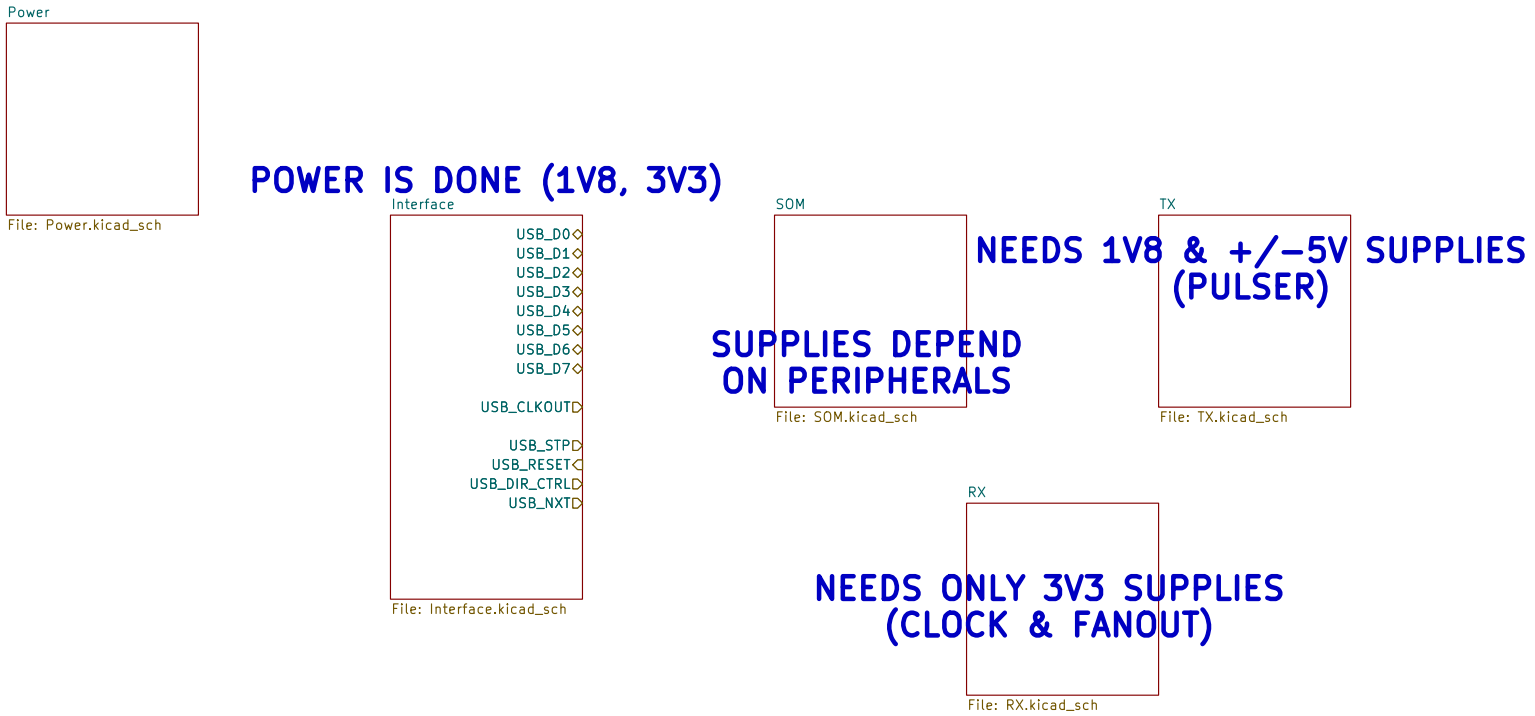
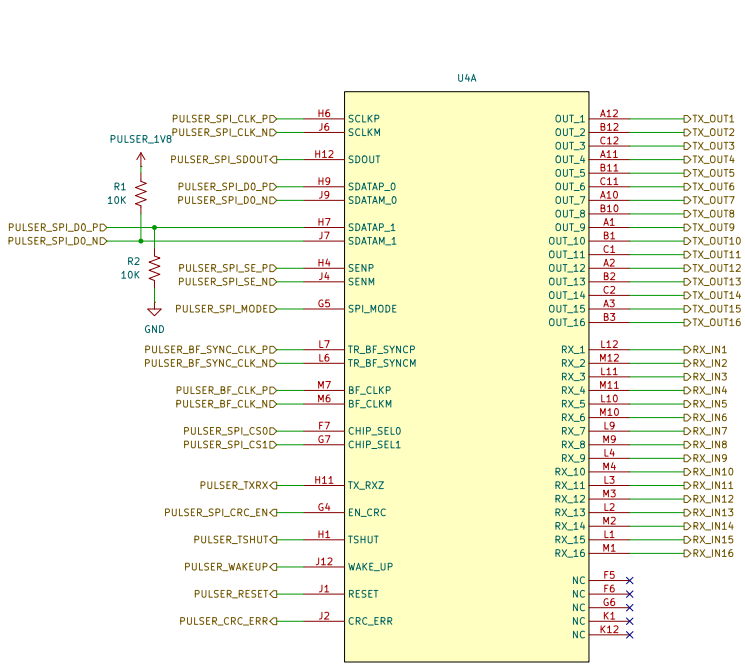
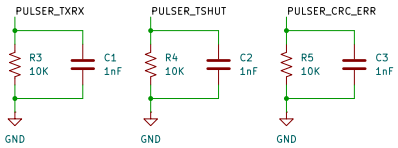
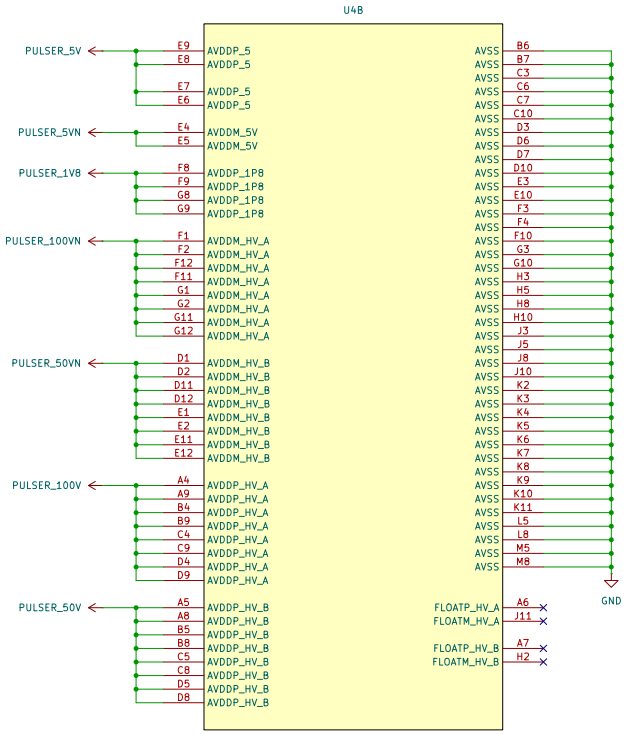


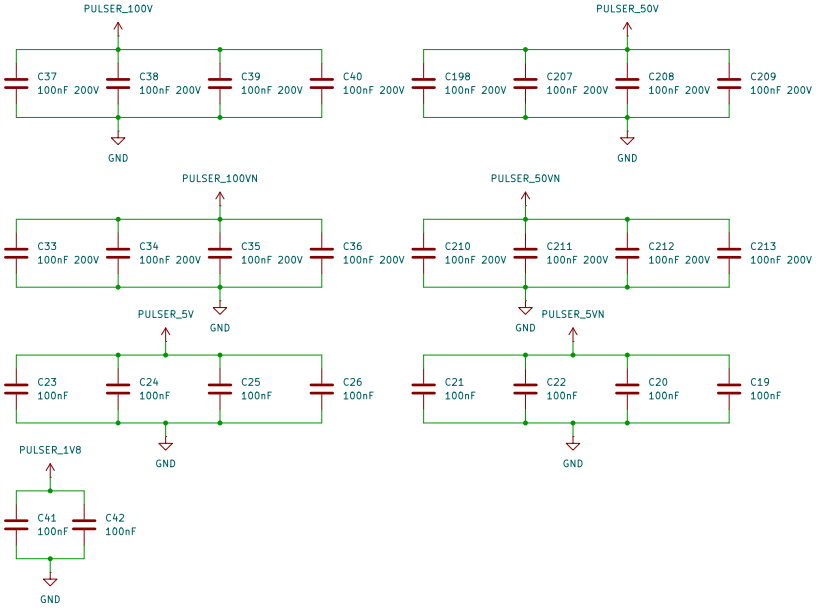
TODO:

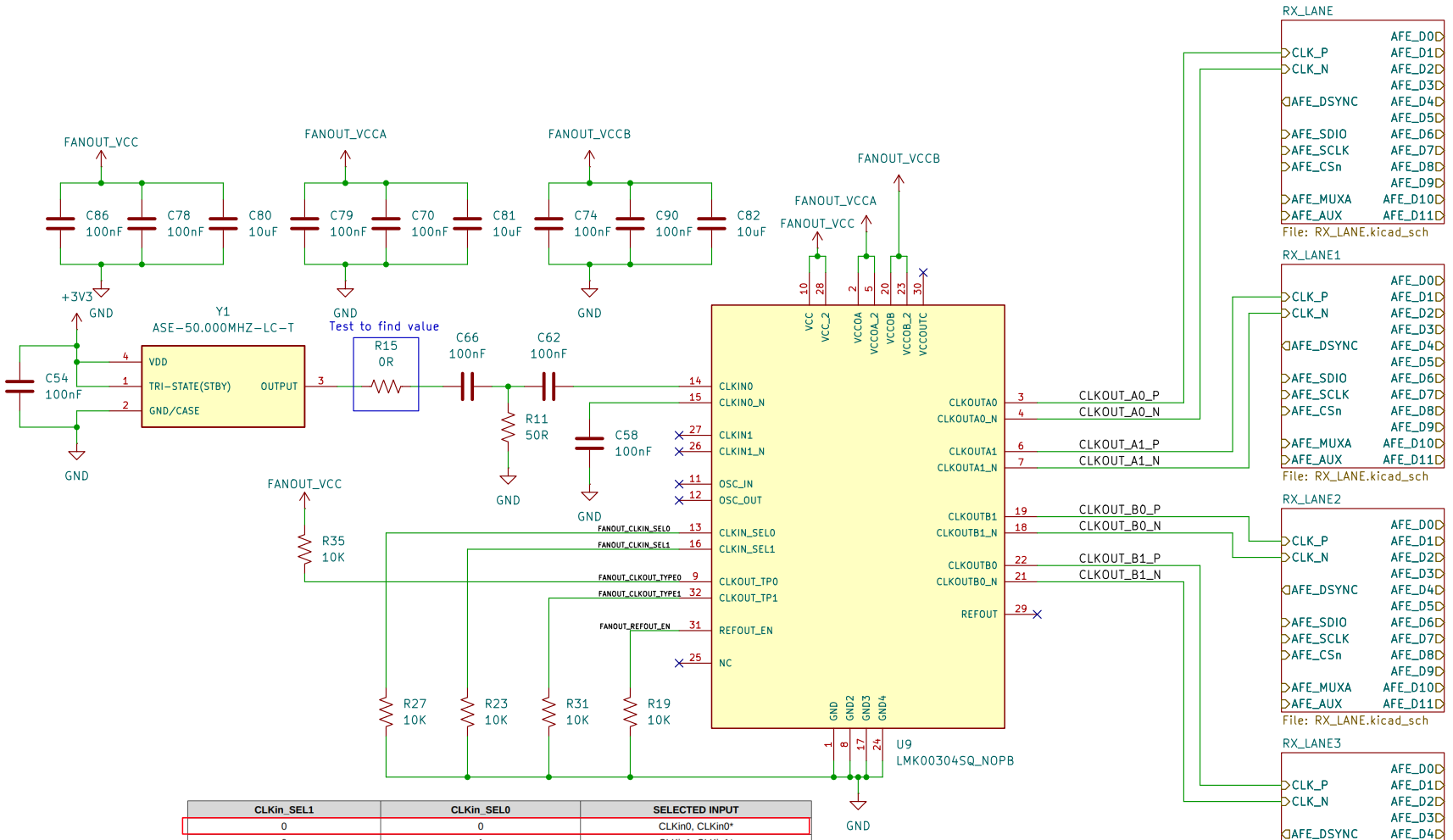
- Add ESD protection to USB & revise, then adjust SOM power
- Add an SD card
- Bind interfaces to the SOM (SPI, SDIO, GPIO)
- Add necessary instances of the LDOs


U4A
PULSER_SPI_CLK_PD
PULSER_SPI_CLK_ND
PULSER_1V8
PULSER_SPI_SDOUT
R1
10K
PULSER_SPI_D0_PD
PULSER_SPI_D0_ND
PULSER_SPI_D0_PD
PULSER_SPI_D0_ND
R2
10K
PULSER_SPI_SE_PD
PULSER_SPI_SE_ND
PULSER_SPI_MODE
GND
PULSER_BF_SYNC_CLK_PD
PULSER_BF_SYNC_CLK_ND
PULSER_BF_CLK_PD
PULSER_BF_CLK_ND
PULSER_SPI_CS0D
PULSER_SPI_CS1D
PULSER_TXRX
PULSER_SPI_CRC_EN
PULSER_TSHUT
PULSER_WAKEUP
PULSER_RESET
PULSER_CRC_ERR
H6
J6
H12
H9
J9
H7
J7
H4
J4
G5
L7
L6
M7
M6
F7
G7
H11
G4
H1
J12
J1
J2
SCLKP
SCLKM
SDOUT
SDATAP_0
SDATAM_0
SDATAP_1
SDATAM_1
SENP
SENM
SPI_MODE
TR_BF_SYNCP
TR_BF_SYNCM
BF_CLKP
BF_CLKM
CHIP_SEL0
CHIP_SEL1
TX_RXZ
EN_CRC
TSHUT
WAKE_UP
RESET
CRC_ERR
OUT_1 A12 TX_OUT1
OUT_2 B12 TX_OUT2
OUT_3 C12 TX_OUT3
OUT_4 A11 TX_OUT4
OUT_5 B11 TX_OUT5
OUT_6 C11 TX_OUT6
OUT_7 A10 TX_OUT7
OUT_8 B10 TX_OUT8
OUT_9 A1 TX_OUT9
OUT_10 B1 TX_OUT10
OUT_11 C1 TX_OUT11
OUT_12 A2 TX_OUT12
OUT_13 B2 TX_OUT13
OUT_14 C2 TX_OUT14
OUT_15 A3 TX_OUT15
OUT_16 B3 TX_OUT16
RX_1 L12 RX_IN1
RX_2 M12 RX_IN2
RX_3 L11 RX_IN3
RX_4 M11 RX_IN4
RX_5 L10 RX_IN5
RX_6 M10 RX_IN6
RX_7 L9 RX_IN7
RX_8 M9 RX_IN8
RX_9 L4 RX_IN9
RX_10 M4 RX_IN10
RX_11 L3 RX_IN11
RX_12 M3 RX_IN12
RX_13 L2 RX_IN13
RX_14 M2 RX_IN14
RX_15 L1 RX_IN15
RX_16 M1 RX_IN16
NC F5
NC F6
NC G6
NC K1
NC K12


PULSER_TXRX
R3
10K
C1
1nF
GND
PULSER_TSHUT
R4
10K
C2
1nF
GND
PULSER_CRC_ERR
R5
10K
C3
1nF
GND

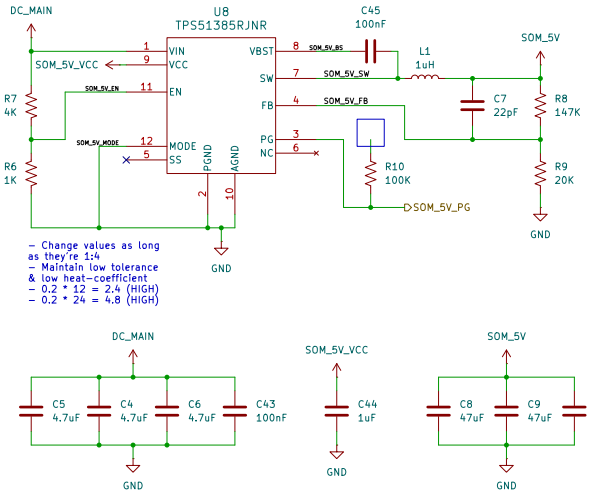

U4B
PULSER_5V
PULSER_5VN
PULSER_1V8
PULSER_100VN
PULSER_50VN
PULSER_100V
PULSER_50V
E9 AVDDP_5
E8 AVDDP_5
E7 AVDDP_5
E6 AVDDP_5
E4 AVDDM_5V
E5 AVDDM_5V
F8 AVDDP_1P8
F9 AVDDP_1P8
G8 AVDDP_1P8
G9 AVDDP_1P8
F1 AVDDM_HV_A
F2 AVDDM_HV_A
F12 AVDDM_HV_A
F11 AVDDM_HV_A
G1 AVDDM_HV_A
G2 AVDDM_HV_A
G11 AVDDM_HV_A
G12 AVDDM_HV_A
D1 AVDDM_HV_B
D2 AVDDM_HV_B
D11 AVDDM_HV_B
D12 AVDDM_HV_B
E1 AVDDM_HV_B
E2 AVDDM_HV_B
E11 AVDDM_HV_B
E12 AVDDM_HV_B
A4 AVDDP_HV_A
A9 AVDDP_HV_A
B4 AVDDP_HV_A
B9 AVDDP_HV_A
C4 AVDDP_HV_A
C9 AVDDP_HV_A
D4 AVDDP_HV_A
D9 AVDDP_HV_A
A5 AVDDP_HV_B
A8 AVDDP_HV_B
B5 AVDDP_HV_B
B8 AVDDP_HV_B
C5 AVDDP_HV_B
C8 AVDDP_HV_B
D5 AVDDP_HV_B
D8 AVDDP_HV_B
AVSS B6
AVSS B7
AVSS C3
AVSS C6
AVSS C7
AVSS C10
AVSS D3
AVSS D6
AVSS D7
AVSS D10
AVSS E3
AVSS E10
AVSS F3
AVSS F4
AVSS F10
AVSS G3
AVSS G10
AVSS H3
AVSS H5
AVSS H8
AVSS H10
AVSS J3
AVSS J5
AVSS J8
AVSS J10
AVSS K2
AVSS K3
AVSS K4
AVSS K5
AVSS K6
AVSS K7
AVSS K8
AVSS K9
AVSS K10
AVSS K11
AVSS L5
AVSS L8
AVSS M5
AVSS M8
FLOATP_HV_A A6
FLOATM_HV_A J11
FLOATP_HV_B A7
FLOATM_HV_B H2
GND


PULSER_100V
C37 100nF 200V
C38 100nF 200V
C39 100nF 200V
C40 100nF 200V
GND
PULSER_50V
C198 100nF 200V
C207 100nF 200V
C208 100nF 200V
C209 100nF 200V
GND
PULSER_100VN
C33 100nF 200V
C34 100nF 200V
C35 100nF 200V
C36 100nF 200V
PULSER_50VN
C210 100nF 200V
C211 100nF 200V
C212 100nF 200V
C213 100nF 200V
PULSER_5V
C23 100nF
C24 100nF
C25 100nF
C26 100nF
GND
PULSER_5VN
C21 100nF
C22 100nF
C20 100nF
C19 100nF
GND
PULSER_1V8
C41 100nF
C42 100nF
GND



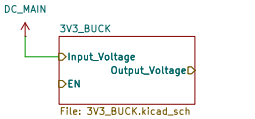
CLKin_SEL1	CLKin_SEL0	SELECTED INPUT
0	0	CLKin0, CLKin0*
0	1	CLKin1, CLKin1*
1	X	OSCin

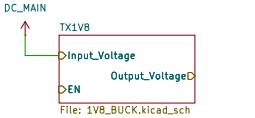
CLKout_ TYPE1	CLKout_ TYPE0	CLKoutX BUFFER TYPE (BANK A and B)
0	0	LVPECL
0	1	LVDS
1	0	HCSL
1	1	Disabled (Hi-Z)

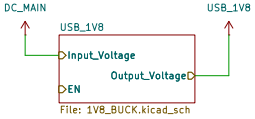


WeBench Designer Link

Power Rail Name	Supported Voltage Range	Maximum Current	Description
V_{CC_SOM}	5V (4.75V – 5.25V) 50 mV p-p maximum noise	4A	Main power input to the SOM. Supplies power to on-board power regulators. Total of 4A across connectors. SOM will consume a maximum of 3A.
V_{CC_BATT}	1.20 – 1.50V	[illegible]	Nominal battery input for the RTC
V_{CCO_HPA}	1.00V – 1.80V	1.0A	Voltage rail for HP3IO bank 66
V_{CCO_HPB}	1.00V – 1.80V	1.0A	Voltage rail for HP3IO bank 65
V_{CCO_HPC}	1.00V – 1.80V	1.0A	Voltage rail for HP3IO bank 64
V_{CCO_HDA}	1.20V – 3.30V	1.0A	Voltage rail for HDIO bank 45
V_{CCO_HDB}	1.20V – 3.30V	1.0A	Voltage rail for HDIO bank 43
V_{CCO_HDC}	1.20V – 3.30V	1.0A	Voltage rail for HDIO bank 44







Duty Cycle

$D_{max} = \frac{V_{out} + V_d}{V_{inmin} + V_{out} + V_d}$ = (100+0.7) / (11+100+0.7) = 90%

$D_{max} = \frac{V_{out} + V_d}{V_{inmin} + V_{out} + V_d}$ = (50+0.7) / (11+50+0.7) = 82%

F_switching = 150 KHz

Inductor

$L1 = L2 = L3 = \frac{V_{inmin} \times V_{inmax} \times D_{max}}{2 \times I_{out} \times V_{out} \times 0.4 \times f_{SW}}$ = (11*25*0.9) / (2*0.3*100*0.4*150000) = 68uH

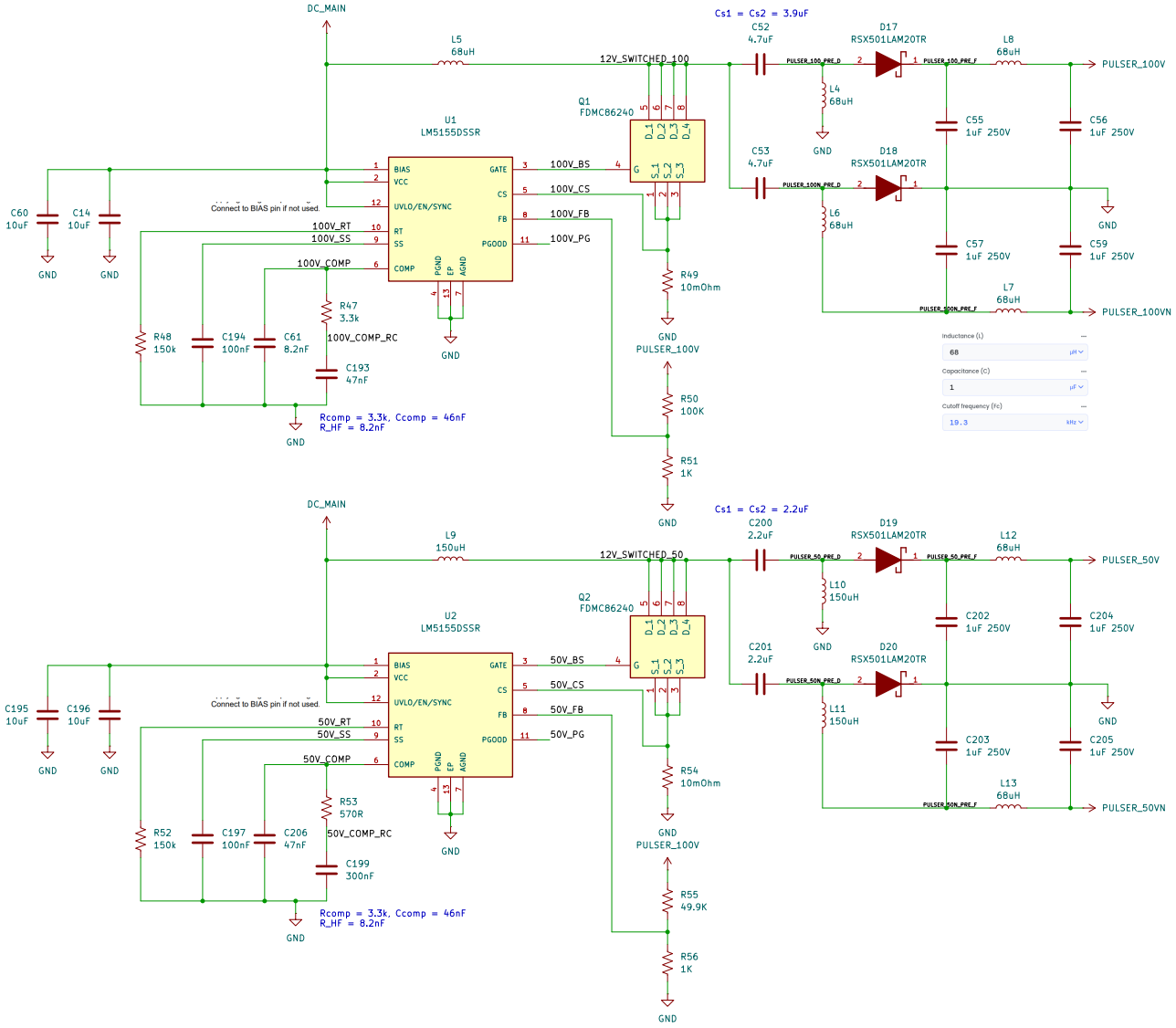
n = 0.88

$I(L1_{peak}) = \frac{I_{out} \times (V_{out} + V_d) \times \left(1 + \frac{0.4}{2}\right)}{V_{inmin} \times n}$ = (0.3*100.7*1.2) / (11*0.88) = 3.75A

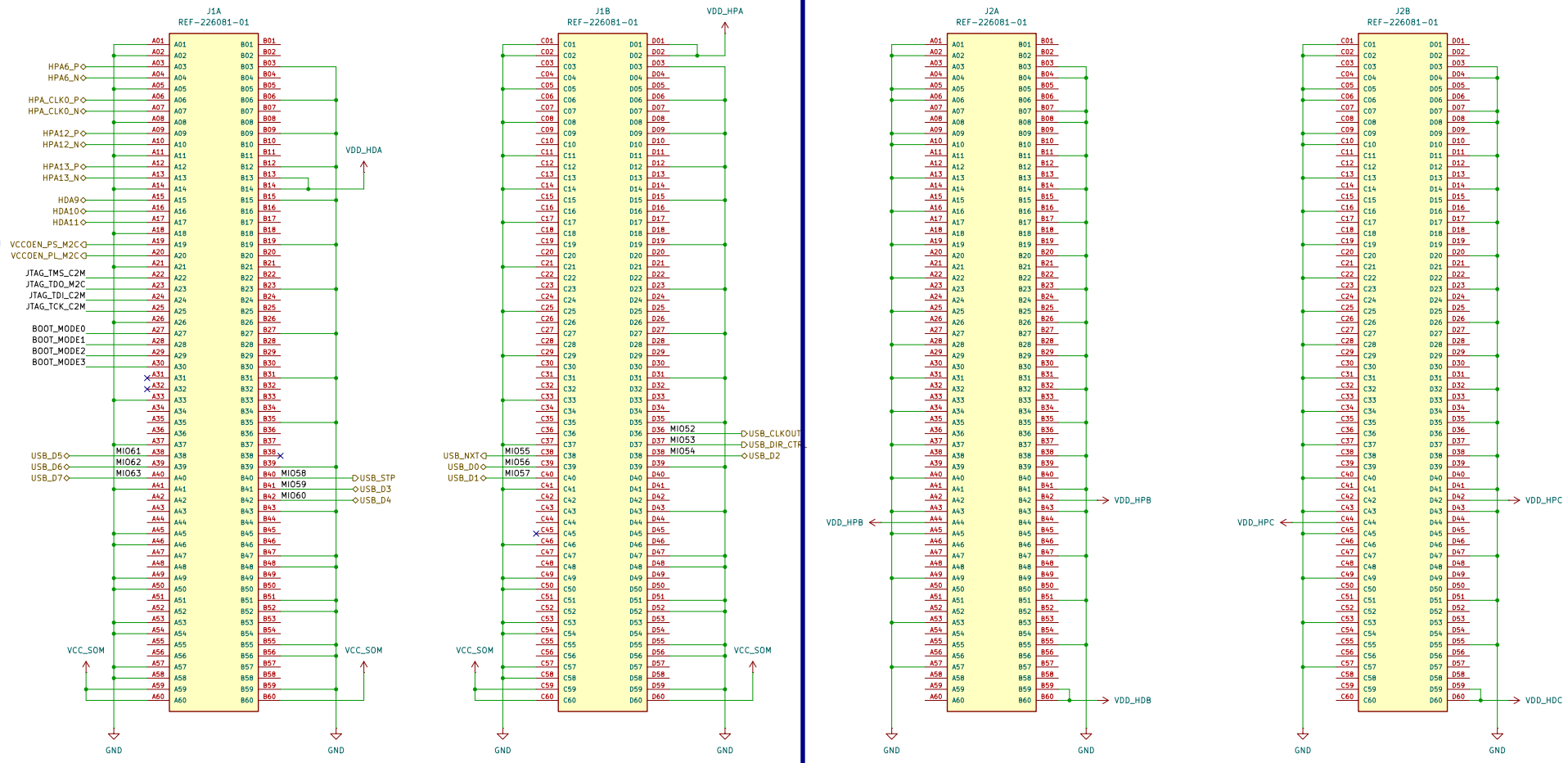
$I(L2_{peak}) = I_{out} \times \left(1 + \frac{0.4}{2}\right)$ = 0.3 * 1.2 = 0.36A

$R_T = \frac{2.21 \times 10^{10}}{f_{RT(TYPICAL)}} - 955$ = 146k (approx. 150k)

$V_{LOAD} = V_{REF} \times \left(\frac{R_{FBT}}{R_{FBB}} + 1\right)$ R_fbt = 99k/49 (approx. 100k/49.9)
R_fbb = 1k

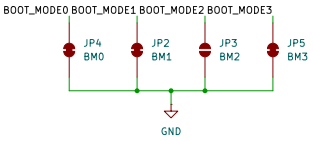


VCC_BATT	PS BBRAM and real-time clock (RTC) supply voltage, requires external battery. Connect to GND when battery is not used.

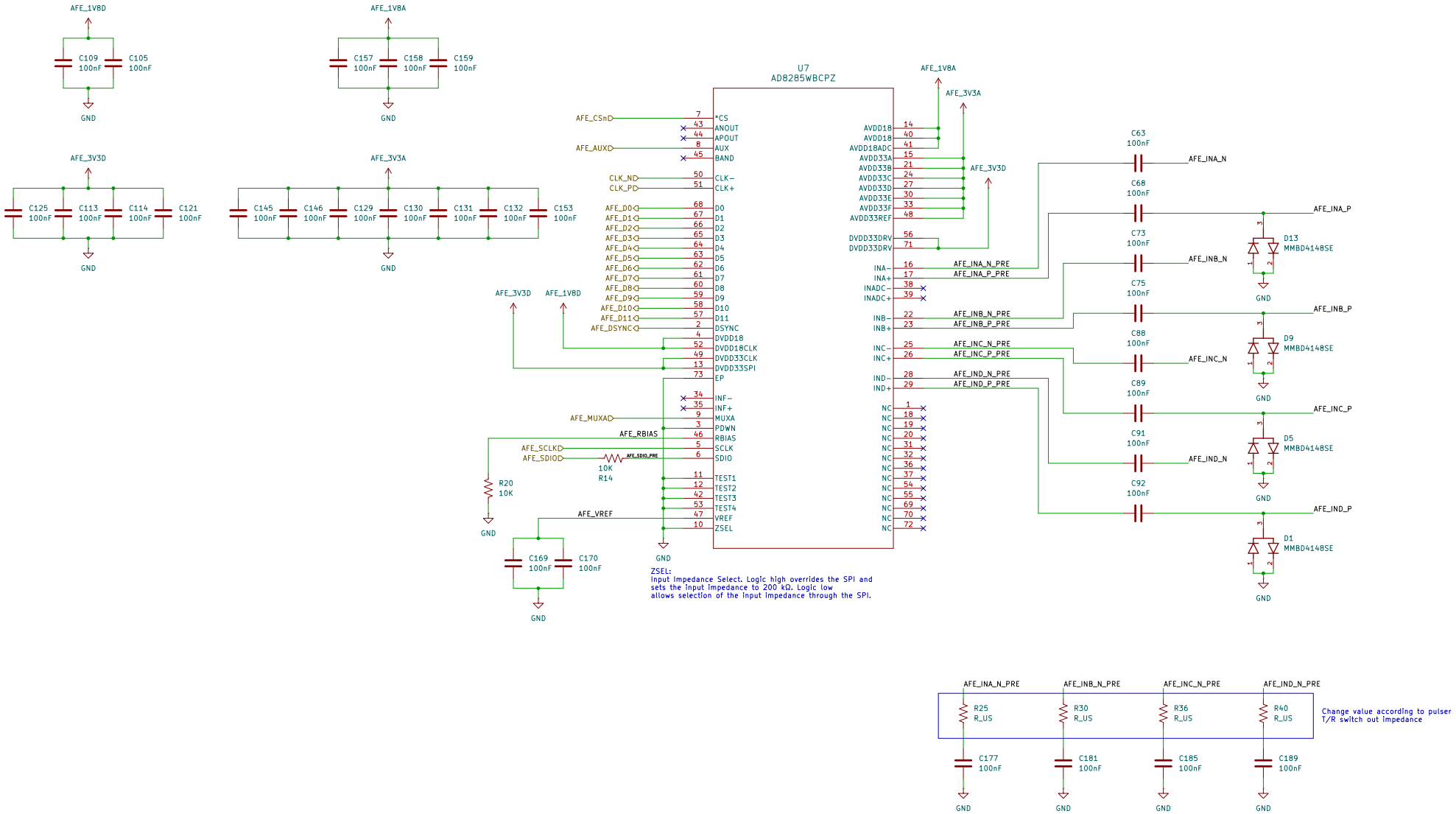


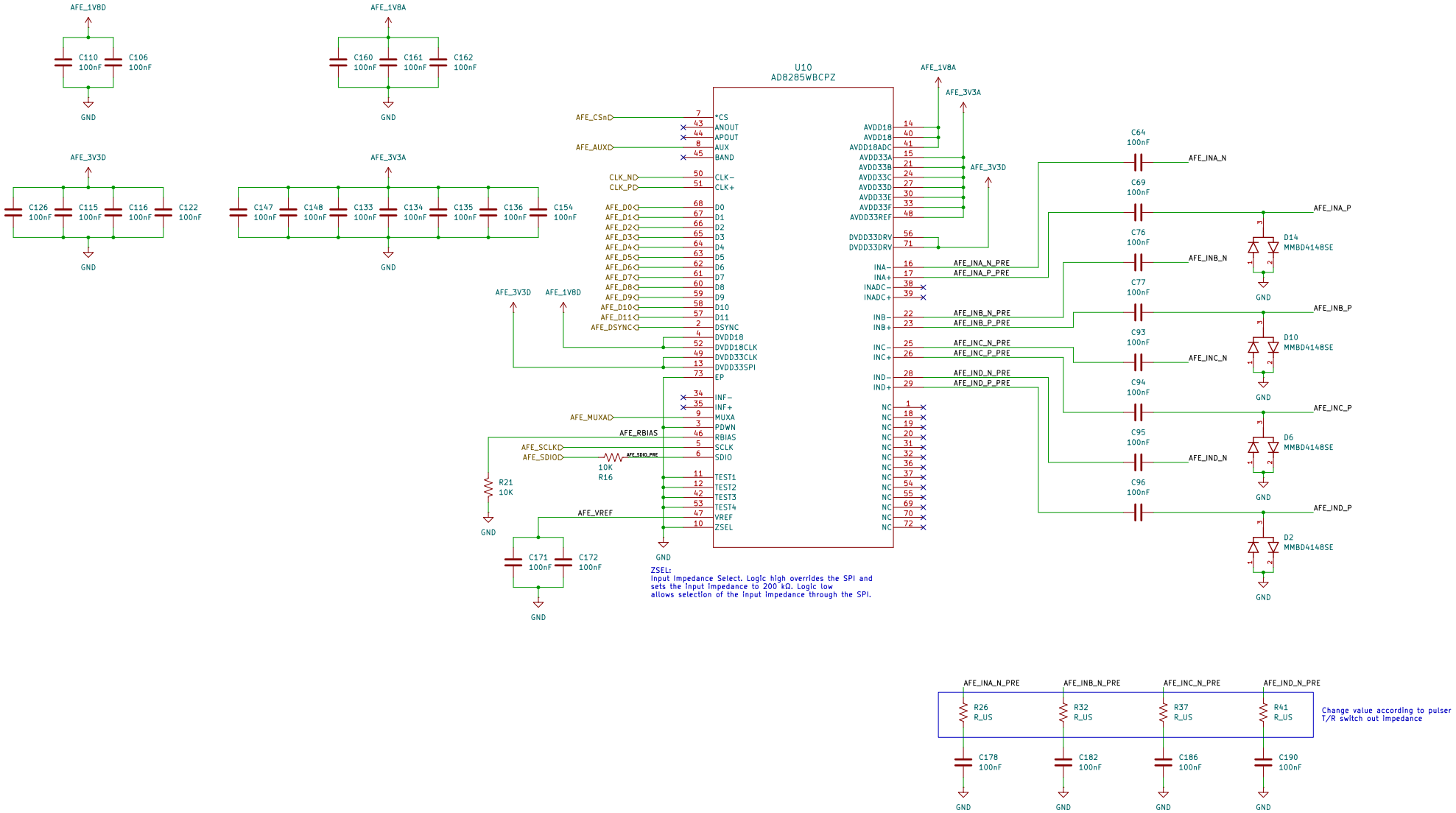
SOM240_1

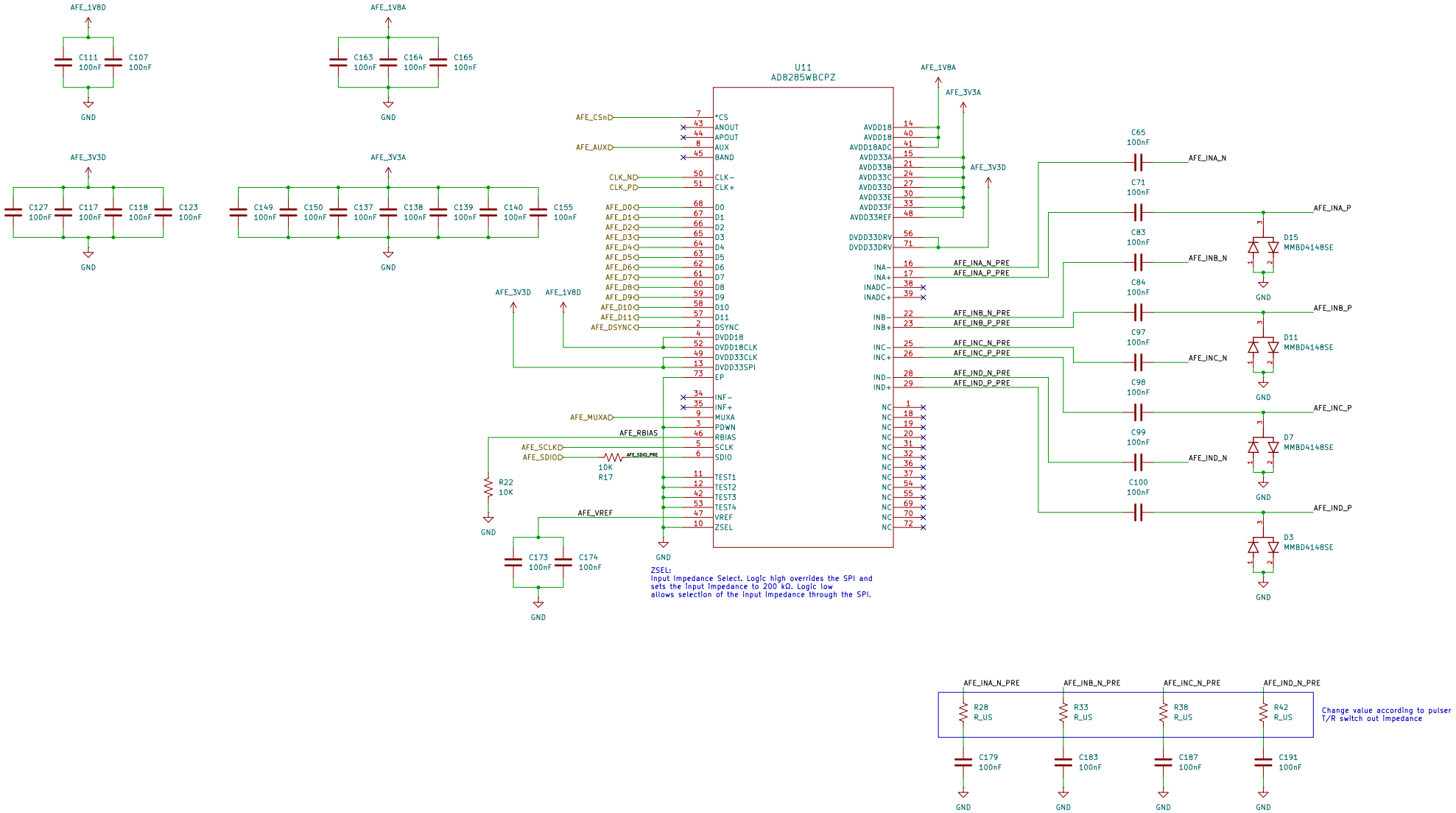
SOM240_2

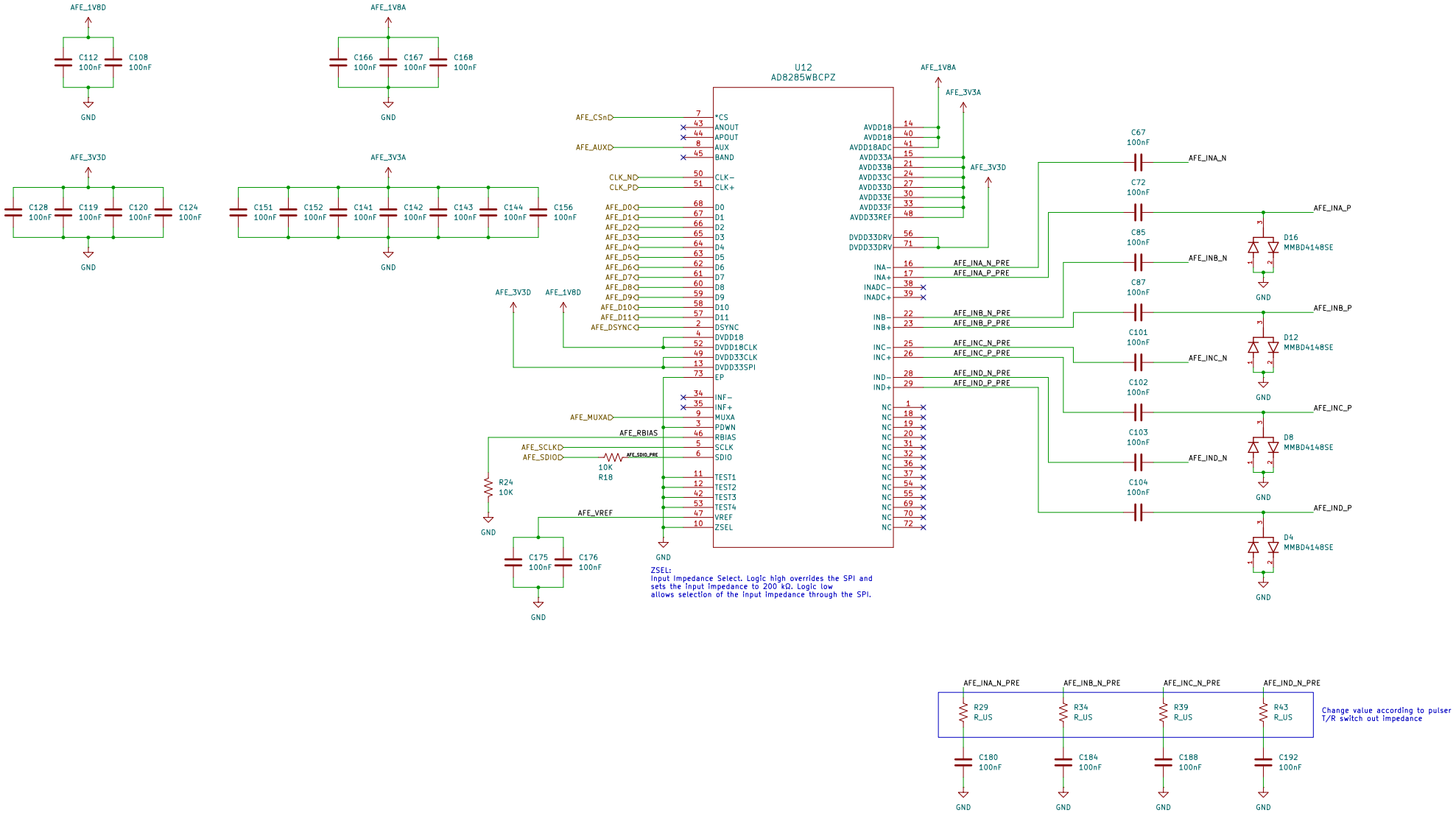


Boot Mode	PS_Mode Pins[3:0]	Physical Pin Location
Quad-SPI (32 bit)	0010	MIO[5:0]
eMMC	0110	MIO[22:13]


AFE_1V8D
C109 100nF
C105 100nF
GND
AFE_1V8A
C157 100nF
C158 100nF
C159 100nF
GND
AFE_3V3D
C125 100nF
C113 100nF
C114 100nF
C121 100nF
GND
AFE_3V3A
C145 100nF
C146 100nF
C129 100nF
C130 100nF
C131 100nF
C132 100nF
C153 100nF
GND
U7
AD8285WBCPZ
AFE_CSn
AFE_AUX
CLK_ND
CLK_PD
AFE_D0
AFE_D1
AFE_D2
AFE_D3
AFE_D4
AFE_D5
AFE_D6
AFE_D7
AFE_D8
AFE_D9
AFE_D10
AFE_D11
AFE_DSYNC
*CS
ANOUT
APOUT
AUX
BAND
CLK−
CLK+
D0
D1
D2
D3
D4
D5
D6
D7
D8
D9
D10
D11
DSYNC
DVDD18
DVDD18CLK
DVDD33CLK
DVDD33SPI
EP
INF−
INF+
MUXA
PDWN
RBIAS
SCLK
SDIO
TEST1
TEST2
TEST3
TEST4
VREF
ZSEL
AVDD18
AVDD18
AVDD18ADC
AVDD33A
AVDD33B
AVDD33C
AVDD33D
AVDD33E
AVDD33F
AVDD33REF
DVDD33DRV
DVDD33DRV
INA−
INA+
INADC−
INADC+
INB−
INB+
INC−
INC+
IND−
IND+
NC
AFE_1V8A
AFE_3V3A
AFE_3V3D
AFE_INA_N_PRE
AFE_INA_P_PRE
AFE_INB_N_PRE
AFE_INB_P_PRE
AFE_INC_N_PRE
AFE_INC_P_PRE
AFE_IND_N_PRE
AFE_IND_P_PRE
AFE_MUXA
AFE_RBIAS
AFE_SCLK
AFE_SDIO
AFE_SDIO_PRE
10K
R14
R20
10K
AFE_VREF
C169 100nF
C170 100nF
GND
ZSEL:
Input Impedance Select. Logic high overrides the SPI and sets the input impedance to 200 kΩ. Logic low allows selection of the input impedance through the SPI.
C63 100nF
C68 100nF
C73 100nF
C75 100nF
C88 100nF
C89 100nF
C91 100nF
C92 100nF
AFE_INA_N
AFE_INA_P
AFE_INB_N
AFE_INB_P
AFE_INC_N
AFE_INC_P
AFE_IND_N
AFE_IND_P
D13 MMBD4148SE
D9 MMBD4148SE
D5 MMBD4148SE
D1 MMBD4148SE
GND
AFE_INA_N_PRE
AFE_INB_N_PRE
AFE_INC_N_PRE
AFE_IND_N_PRE
R25 R_US
R30 R_US
R36 R_US
R40 R_US
C177 100nF
C181 100nF
C185 100nF
C189 100nF
GND
Change value according to pulser T/R switch out impedance

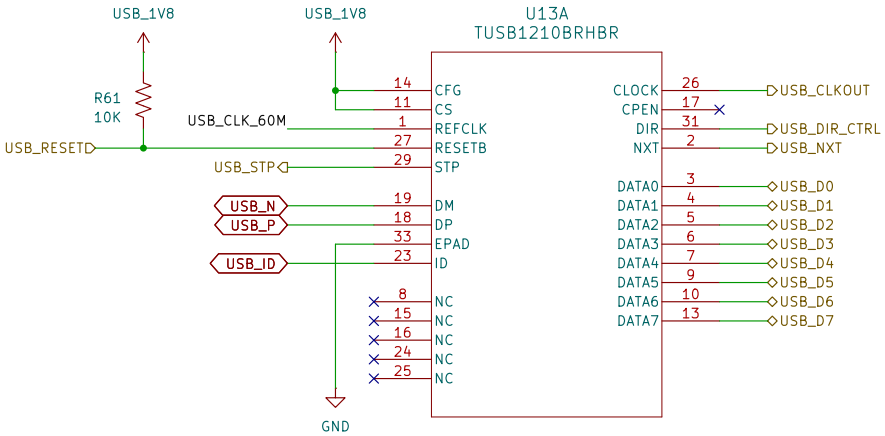

AFE_1V8D
C110 100nF
C106 100nF
GND
AFE_1V8A
C160 100nF
C161 100nF
C162 100nF
GND
AFE_3V3D
C126 100nF
C115 100nF
C116 100nF
C122 100nF
GND
AFE_3V3A
C147 100nF
C148 100nF
C133 100nF
C134 100nF
C135 100nF
C136 100nF
C154 100nF
GND
U10
AD8285WBCPZ
AFE_CSn
AFE_AUXD
CLK_ND
CLK_PD
AFE_D0
AFE_D1
AFE_D2
AFE_D3
AFE_D4
AFE_D5
AFE_D6
AFE_D7
AFE_D8
AFE_D9
AFE_D10
AFE_D11
AFE_DSYNC
*CS
ANOUT
APOUT
AUX
BAND
CLK−
CLK+
D0
D1
D2
D3
D4
D5
D6
D7
D8
D9
D10
D11
DSYNC
DVDD18
DVDD18CLK
DVDD33CLK
DVDD33SPI
EP
INF−
INF+
MUXA
PDWN
RBIAS
SCLK
SDIO
TEST1
TEST2
TEST3
TEST4
VREF
ZSEL
AVDD18
AVDD18
AVDD18ADC
AVDD33A
AVDD33B
AVDD33C
AVDD33D
AVDD33E
AVDD33F
AVDD33REF
DVDD33DRV
DVDD33DRV
INA−
INA+
INADC−
INADC+
INB−
INB+
INC−
INC+
IND−
IND+
NC
AFE_INA_N_PRE
AFE_INA_P_PRE
AFE_INB_N_PRE
AFE_INB_P_PRE
AFE_INC_N_PRE
AFE_INC_P_PRE
AFE_IND_N_PRE
AFE_IND_P_PRE
AFE_MUXAD
AFE_RBIAS
AFE_SCLKD
AFE_SDIOD
AFE_SDIO_PRE
10K
R16
R21
10K
AFE_VREF
C171 100nF
C172 100nF
ZSEL:
Input Impedance Select. Logic high overrides the SPI and sets the input impedance to 200 kΩ. Logic low allows selection of the input impedance through the SPI.
C64 100nF
C69 100nF
C76 100nF
C77 100nF
C93 100nF
C94 100nF
C95 100nF
C96 100nF
AFE_INA_N
AFE_INA_P
AFE_INB_N
AFE_INB_P
AFE_INC_N
AFE_INC_P
AFE_IND_N
AFE_IND_P
D14 MMBD4148SE
D10 MMBD4148SE
D6 MMBD4148SE
D2 MMBD4148SE
R26 R_US
R32 R_US
R37 R_US
R41 R_US
Change value according to pulser T/R switch out impedance
C178 100nF
C182 100nF
C186 100nF
C190 100nF

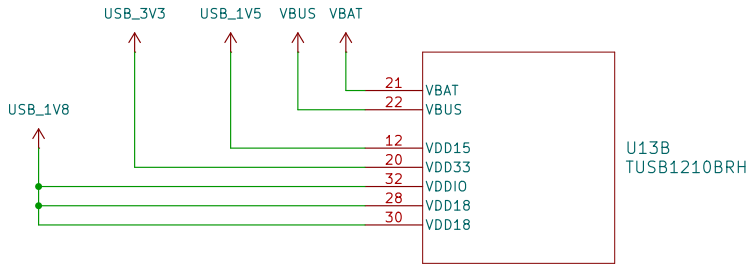

AFE_1V8D
C111 100nF
C107 100nF
GND
AFE_1V8A
C163 100nF
C164 100nF
C165 100nF
GND
AFE_3V3D
C127 100nF
C117 100nF
C118 100nF
C123 100nF
GND
AFE_3V3A
C149 100nF
C150 100nF
C137 100nF
C138 100nF
C139 100nF
C140 100nF
C155 100nF
GND
U11
AD8285WBCPZ
AFE_CSn
AFE_AUXD
CLK_ND
CLK_PD
AFE_D0
AFE_D1
AFE_D2
AFE_D3
AFE_D4
AFE_D5
AFE_D6
AFE_D7
AFE_D8
AFE_D9
AFE_D10
AFE_D11
AFE_DSYNC
*CS
ANOUT
APOUT
AUX
BAND
CLK−
CLK+
D0
D1
D2
D3
D4
D5
D6
D7
D8
D9
D10
D11
DSYNC
DVDD18
DVDD18CLK
DVDD33CLK
DVDD33SPI
EP
INF−
INF+
MUXA
PDWN
RBIAS
SCLK
SDIO
TEST1
TEST2
TEST3
TEST4
VREF
ZSEL
AVDD18
AVDD18
AVDD18ADC
AVDD33A
AVDD33B
AVDD33C
AVDD33D
AVDD33E
AVDD33F
AVDD33REF
DVDD33DRV
DVDD33DRV
INA−
INA+
INADC−
INADC+
INB−
INB+
INC−
INC+
IND−
IND+
NC
AFE_1V8A
AFE_3V3A
AFE_3V3D
AFE_INA_N_PRE
AFE_INA_P_PRE
AFE_INB_N_PRE
AFE_INB_P_PRE
AFE_INC_N_PRE
AFE_INC_P_PRE
AFE_IND_N_PRE
AFE_IND_P_PRE
AFE_MUXA
AFE_RBIAS
AFE_SCLK
AFE_SDIO
AFE_SDIO_PRE
10K
R17
R22
10K
AFE_VREF
C173 100nF
C174 100nF
ZSEL:
Input Impedance Select. Logic high overrides the SPI and sets the input impedance to 200 kΩ. Logic low allows selection of the input impedance through the SPI.
C65 100nF
C71 100nF
C83 100nF
C84 100nF
C97 100nF
C98 100nF
C99 100nF
C100 100nF
AFE_INA_N
AFE_INA_P
AFE_INB_N
AFE_INB_P
AFE_INC_N
AFE_INC_P
AFE_IND_N
AFE_IND_P
D15 MMBD4148SE
D11 MMBD4148SE
D7 MMBD4148SE
D3 MMBD4148SE
AFE_INA_N_PRE
AFE_INB_N_PRE
AFE_INC_N_PRE
AFE_IND_N_PRE
R28 R_US
R33 R_US
R38 R_US
R42 R_US
Change value according to pulser T/R switch out impedance
C179 100nF
C183 100nF
C187 100nF
C191 100nF

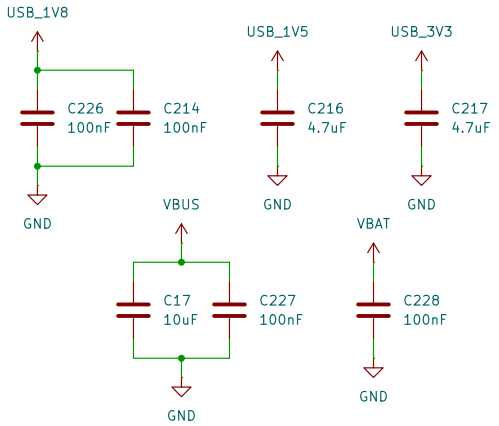

AFE_1V8D
C112 100nF
C108 100nF
GND
AFE_1V8A
C166 100nF
C167 100nF
C168 100nF
GND
AFE_3V3D
C128 100nF
C119 100nF
C120 100nF
C124 100nF
GND
AFE_3V3A
C151 100nF
C152 100nF
C141 100nF
C142 100nF
C143 100nF
C144 100nF
C156 100nF
GND
U12
AD8285WBCPZ
AFE_CSn
AFE_AUX
CLK_ND
CLK_PD
AFE_D0
AFE_D1
AFE_D2
AFE_D3
AFE_D4
AFE_D5
AFE_D6
AFE_D7
AFE_D8
AFE_D9
AFE_D10
AFE_D11
AFE_DSYNC
*CS
ANOUT
APOUT
AUX
BAND
CLK−
CLK+
D0
D1
D2
D3
D4
D5
D6
D7
D8
D9
D10
D11
DSYNC
DVDD18
DVDD18CLK
DVDD33CLK
DVDD33SPI
EP
INF−
INF+
MUXA
PDWN
RBIAS
SCLK
SDIO
TEST1
TEST2
TEST3
TEST4
VREF
ZSEL
AVDD18
AVDD18
AVDD18ADC
AVDD33A
AVDD33B
AVDD33C
AVDD33D
AVDD33E
AVDD33F
AVDD33REF
DVDD33DRV
DVDD33DRV
INA−
INA+
INADC−
INADC+
INB−
INB+
INC−
INC+
IND−
IND+
NC
AFE_MUXA
AFE_RBIAS
AFE_SCLK
AFE_SDIO
AFE_SDIO_PRE
10K
R18
R24
10K
AFE_VREF
C175 100nF
C176 100nF
ZSEL:
Input Impedance Select. Logic high overrides the SPI and sets the input impedance to 200 kΩ. Logic low allows selection of the input impedance through the SPI.
AFE_1V8A
AFE_3V3A
AFE_3V3D
AFE_INA_N_PRE
AFE_INA_P_PRE
AFE_INB_N_PRE
AFE_INB_P_PRE
AFE_INC_N_PRE
AFE_INC_P_PRE
AFE_IND_N_PRE
AFE_IND_P_PRE
C67 100nF
C72 100nF
C85 100nF
C87 100nF
C101 100nF
C102 100nF
C103 100nF
C104 100nF
AFE_INA_N
AFE_INA_P
AFE_INB_N
AFE_INB_P
AFE_INC_N
AFE_INC_P
AFE_IND_N
AFE_IND_P
D16 MMBD4148SE
D12 MMBD4148SE
D8 MMBD4148SE
D4 MMBD4148SE
GND
AFE_INA_N_PRE
AFE_INB_N_PRE
AFE_INC_N_PRE
AFE_IND_N_PRE
R29 R_US
R34 R_US
R39 R_US
R43 R_US
Change value according to pulser T/R switch out impedance
C180 100nF
C184 100nF
C188 100nF
C192 100nF
GND

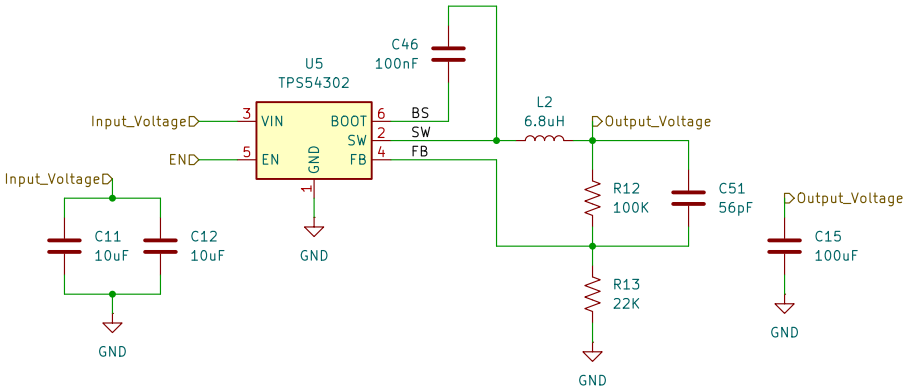
Check AN1310 for layout guidelines

CFG	14	D	I	V_{DDIO}	REFCLK clock frequency configuration pin. Two frequencies are supported: 19.2 MHz when 0, or 26 MHz when 1.
CS	11	D	I	V_{DDIO}	Active-high chip select pin. When low the IC is in power down and ULPI bus is tri-stated. When high normal operation. Tie to V_{DDIO} if unused.

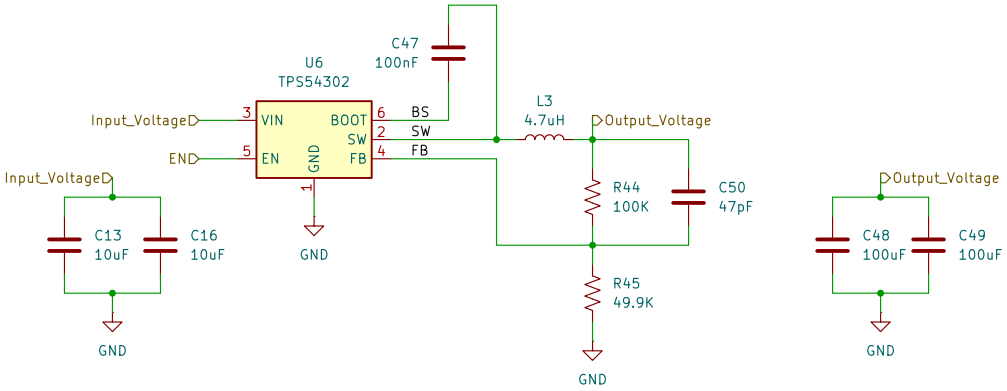




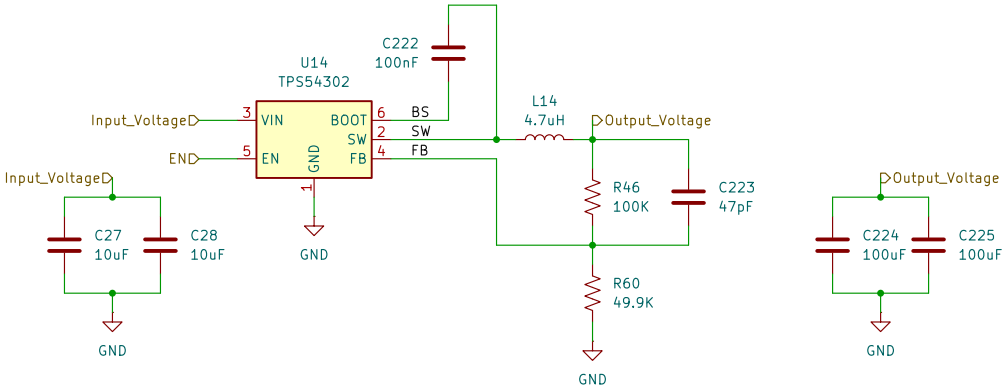




WEBENCH DESIGNER LINK (3.3V)



WEBENCH DESIGNER LINK (1.8V)



WEBENCH DESIGNER LINK (1.8V)